Chapter 10. Separation Theorems in the Plane

Jordan separation, invariance of domain, Jordan curve theorem, graphs in the plane, winding numbers

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We apply the fundamental group to the topology of the plane. It is convenient to work in $S^2 = \mathbb{R}^2 \cup \{\infty\}$, where $S^2 - \{\text{point}\} \cong \mathbb{R}^2$. An **arc** is a space homeomorphic to $[0,1]$ and a **simple closed curve** one homeomorphic to S^1. A set C **separates** X if $X - C$ is not connected.

10.1 The Jordan separation theorem

Lemma 10.1 (Nulhomotopy lemma)

Let $a, b \in S^2$ and A a compact space. If $f : A \to S^2 - \{a, b\}$ is continuous and a, b lie in the same component of $S^2 - f(A)$, then f is nulhomotopic as a map into $S^2 - \{a, b\}$.

Idea. Put a at ∞; then $S^2 - \{a, b\} \cong \mathbb{R}^2 - \{0\}$, which deformation retracts to S^1. Since $f(A)$ is compact and b can be joined to ∞ avoiding $f(A)$, the map $x \mapsto (f(x) - b)/\|f(x) - b\|$ is homotopic to a map into a small arc, by moving b along a path to a point far away. ■

The basic tool is the following consequence of covering space theory.

Theorem 10.2 (Two-component lemma)

Let $X = U \cup V$ with U, V open and $U \cap V = A \cup B$, where A, B are disjoint, open and path connected. Let α be a path in U and β a path in V, both from $a \in A$ to $b \in B$. Then the loop $\alpha * \bar{\beta}$ generates an infinite cyclic subgroup of $\pi_1(X, a)$; in particular $\pi_1(X) \neq 0$.

Idea. Build a covering space E of X by gluing countably many copies $U \times \{2n\}$ and $V \times \{2n+1\}$, identifying A in $U \times \{2n\}$ with A in $V \times \{2n-1\}$ and B in $U \times \{2n\}$ with B in $V \times \{2n+1\}$. The loop $(\alpha * \bar{\beta})^m$ lifts to a path from level 0 to level $2m$, so it is not trivial for $m \neq 0$. ■

Theorem 10.3

(a) An arc does not separate S^2. (b) The Jordan separation theorem: a simple closed curve C separates S^2.

Idea. (a) If an arc D separated a from b, split D into two halves; by induction (bisecting the half that still separates), we get nested subarcs shrinking to a point whose complement-inclusions π_1 would all be nontrivial, contradicting that S^2 minus a point is simply connected. The key tool is: if $X = U \cup V$ with $U \cap V$ having exactly two path components, then $\pi_1(X)$ is nontrivial (a loop crossing between the components cannot be contracted), applied in the contrapositive. (b) Write $C = A_1 \cup A_2$ as two arcs meeting at two points. If $S^2 - C$ were connected, apply the same principle to $U = S^2 - A_1$, $V = S^2 - A_2$: $U \cup V = S^2 - \{p, q\}$ has infinite cyclic π_1, while by (a) and the nulhomotopy lemma the relevant loop would be trivial — a contradiction. ■

More generally, if D_1, D_2 are closed connected subsets of S^2 meeting in exactly two points and neither separates S^2, then $D_1 \cup D_2$ does.

10.2 Invariance of domain

Theorem 10.4 (Invariance of domain, $n = 2$)

If $U \subset \mathbb{R}^2$ is open and $f : U \to \mathbb{R}^2$ is continuous and injective, then $f(U)$ is open and f is an imbedding.

Idea. For a small closed disc B in U with boundary circle S, $f(S)$ is a simple closed curve and $f(B)$ is an arc-like non-separating set; $f(\operatorname{Int} B)$ is a connected set disjoint from $f(S)$ and must be one of the components of $\mathbb{R}^2 - f(S)$ (using the separation theorem and a π_1 argument showing the bounded component is filled), hence open. ■

Consequently $\mathbb{R}^2$ and $\mathbb{R}^n$ $(n \neq 2)$ are not homeomorphic, and a 2-manifold cannot be an n-manifold for $n \neq 2$.

10.3 The Jordan curve theorem

Theorem 10.5 (Jordan curve theorem)

If C is a simple closed curve in S^2, then $S^2 - C$ has exactly two components, and C is the common boundary of each.

Idea. At least two components: Theorem 10.3. At most two: if U_1, U_2, U_3 were three components, using arcs and the fact that $\pi_1(S^2 - \{p, q\}) \cong \mathbb{Z}$ one produces a loop that is both nontrivial and trivial. Boundary: if C minus an open arc failed to be in the boundary of a component, the complementary arc would separate S^2, contradicting Theorem 10.3(a). ■

In $\mathbb{R}^2$ the two components are one bounded (the inside) and one unbounded. The **Schoenflies theorem** (not proved here) says the inside together with C is a closed disc. Both results fail in higher dimensions in the stronger form (Alexander horned sphere), though the separation statement persists.

10.4 Imbedding graphs in the plane

A (finite) **linear graph** is a space that is the union of finitely many arcs meeting at most at endpoints. A **theta space** is the union of three arcs with common endpoints.

Lemma 10.6

If $X \subset S^2$ is a theta space, $S^2 - X$ has exactly three components, with boundaries the three simple closed curves in X.

Theorem 10.7

The **utilities graph** $K_{3,3}$ (three houses joined to three utilities) cannot be imbedded in the plane. Neither can the complete graph K_5.

Idea. $K_{3,3}$ contains a theta space whose complement has three faces; the remaining vertex lies in one face whose boundary misses one of the three vertices it must be joined to, and the joining arc would have to cross the theta space. The argument for K_5 is similar. ■

(Kuratowski's theorem says these are the only obstructions.) Euler's formula $V - E + F = 2$ for connected planar graphs gives an alternative argument.

10.5 The winding number

For a loop $f : S^1 \to \mathbb{R}^2 - \{a\}$, the **winding number** $n(f, a)$ is the integer k such that $x \mapsto \frac{f(x)-a}{\|f(x)-a\|}$ is homotopic to $z \mapsto z^k$; equivalently the degree of this map, computed by lifting to $\mathbb{R}$.

Theorem 10.8

$n(f, a)$ is constant as a varies in a component of $\mathbb{R}^2 - f(S^1)$, and is 0 on the unbounded component. If f is a simple closed curve, then $n(f, a) = \pm 1$ for a inside and 0 outside.

Idea. Continuity in a and integrality give constancy on components; for a far away the normalized map has small image. For a simple closed curve, deform C near a point to a small circle around a and use the Jordan curve theorem to localize. ■

10.6 The Cauchy integral formula

For a piecewise smooth loop f in $\mathbb{C} - \{a\}$, $n(f, a) = \frac{1}{2\pi i} \int_f \frac{dz}{z-a}$: the integral computes the change of a continuous branch of $\log(z - a)$ along f, which is $2\pi i$ times the change of the lifted argument. Combined with Cauchy's theorem for simply connected domains (a holomorphic g on a simply connected region has $\int_f g = 0$ for every closed curve, by homotopy invariance of path integrals), this gives the Cauchy integral formula

$$n(f, a)\, g(a) = \frac{1}{2\pi i} \int_f \frac{g(z)}{z - a} dz,$$

and for a simple closed curve oriented counterclockwise $n(f, a) = 1$ at interior points a.